

Θ -H Control: From Explosive Divergence to Quasi-Periodic Order in Quartic Nonlinear Systems

Marconi Kolm Madrid
madrid@unicamp.br

Departamento de Sistemas de Energia
Faculdade de Engenharia Elétrica e de Computação
Universidade Estadual de Campinas (UNICAMP)

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Abstract

This paper presents Θ -H, a hybrid adaptive controller capable of transforming explosive dynamical systems into stable quasi-periodic behavior. The application is demonstrated to a system with quartic nonlinearity (x^4), which in open loop diverges to infinity in finite time – a more severe behavior than chaos. Under Θ -H action, the explosive system is stabilized and organized into a quasi-periodic attractor with Lyapunov exponent $\lambda = 0.0014 \approx 0$, characterized by a torus in the Poincaré map and the presence of multiple coexisting orbits. The controller operates without prior knowledge of the model, learning the dynamics through safe autonomous exploration. Experimental results show a spontaneous transition between low and high amplitude orbits after approximately 4300 crossings, maintaining the same fundamental frequency. Θ -H represents a new paradigm for controlling systems with severe nonlinearities, offering an alternative to traditional mathematical modeling.

Keywords: Θ -H Control, Quartic Nonlinearity, Quasi-Periodic Systems, Explosive Dynamics, Lyapunov Exponent.

1 Introduction

1.1 Context and Motivation

Nonlinear dynamical systems exhibit complex behaviors such as bifurcations, chaos, and in extreme cases, explosive divergence [1–3]. While chaos has been extensively studied, systems with higher-order nonlin-

earities (quartic, quintic) remain less explored due to their tendency toward catastrophic instability [4].

The Duffing oscillator (x^3) [1] is a paradigmatic example of chaos. However, systems with x^4 nonlinearity – here termed quartic oscillators – exhibit qualitatively different behavior: explosive divergence in open loop. Controlling such systems is a significantly greater challenge, as there is no natural attractor to tame.

Recent advances in data-driven control have introduced model-free approaches based on machine learning [5], but these require extensive offline training. Adaptive methods for high-order nonlinear systems [6] achieve fixed-time convergence at the cost of high computational complexity. Chen et al. [7] demonstrated the generation of quasi-periodic and chaotic attractors via switching schemes, though without autonomous learning capabilities. The present work fills this gap with Theta-H, a controller that combines computational simplicity, real-time learning, and the unique ability to handle explosive systems.

1.2 Contributions

This paper presents the following contributions:

1. A novel hybrid adaptive controller, called Θ -H, combining state feedback with BK memory and autonomous learning.
2. The first demonstration of controlling an explosive quartic system, transforming its dynamics into stable quasi-periodic behavior.

3. Discovery of a rare phenomenon: multiple coexisting quasi-periodic orbits on the same torus, with spontaneous transition between low and high amplitude regimes.
4. Experimental evidence of Lyapunov exponent $\lambda = 0.0014 \approx 0$, confirming the quasi-periodic nature of the controlled system.

1.3 Paper Organization

Section 2 describes the Θ -H architecture. Section 3 details the internal mechanics, including the BK memory, safe autonomous exploration. Section 4 presents the quartic system. Section 5 shows experimental results. Section 6 discusses implications. Section 7 concludes.

2 Θ -H Architecture

2.1 General Structure

Θ -H is defined by:

$$u(t) = -K \cdot x(t) + u_{\Theta}(t) + u_{BK}(t) \quad (1)$$

where:

- $K = [K_1, K_2]$ are fixed gains ($K_1 = 8$, $K_2 = 6$ in this work)
- $u_{\Theta}(t)$ is the adaptive (learned) action
- $u_{BK}(t)$ is the BK memory (low-pass filter)

2.2 Θ Action

The Θ action is given by an adaptive PID controller:

$$u_{\Theta} = K_p e + K_i \int e dt + K_d \dot{e} \quad (2)$$

With gains derived from τ parameters:

$$K_p = \frac{1.2}{\tau_1}, \quad K_i = \frac{0.6}{\tau_1 \tau_2}, \quad K_d = 0.5 \tau_1 \tau_3 \quad (3)$$

In this work, the values $\tau_1 = 0.5$, $\tau_2 = 0.4$, $\tau_3 = 0.1$ are used.

2.3 BK Memory

The BK memory acts as a low-pass filter:

$$u_{BK}(t) = \alpha \cdot u_{BK}(t - dt) + (1 - \alpha) \cdot u_{fb}(t), \quad (4)$$

where $\alpha = 0.98$ is the memory factor, and $u_{fb}(t) = -K \cdot x(t)$.

2.4 Autonomous Learning

Θ -H learns the plant dynamics through:

1. Safe exploration with increasing amplitude impulses
2. Identification of the growth rate λ
3. Autonomous adjustment of the time window

3 Θ -H Internal Mechanics: The Learning Engine

3.1 The Memory-BK Low-Pass Filter

One of the key innovations of Θ -H is the BK memory mechanism:

$$u_{BK}(t) = \alpha u_{BK}(t - dt) + (1 - \alpha) u_{fb}(t), \quad \alpha = 0.98 \quad (5)$$

This first-order low-pass filter introduces a time-delayed feedback that creates an emergent quasi-periodic structure. Without BK ($\alpha = 0$), the controller becomes purely instantaneous and the system remains chaotic or explosive. With BK ($\alpha \approx 1$), the controller gains a memory of its own past actions, creating a self-referential dynamical system.

3.2 Safe Autonomous Exploration of Explosive Plants

Learning the dynamics of an explosive system requires extreme caution. Θ -H employs a hierarchical exploration strategy:

1. Impulse Testing: Small amplitude impulses ($A = 0.01$) are applied. If the response is below a safety threshold ($|y| < 0.1$), amplitude is doubled.

2. Time Window Adaptation: The identification window is automatically shifted later in time ($t_{window} = 1.2, 1.4, 1.6, \dots$) until the estimated growth rate λ stabilizes.
3. Learning Convergence: After approximately 10-15 safe impulses, the system's effective parameters are identified with sufficient accuracy to design the compensator.

3.3 Why Θ -H Transforms Explosive Dynamics into Quasi-Periodicity

The controlled quartic system exhibits an emergent property: spontaneous transitions between low and high amplitude orbits with identical fundamental frequency. This phenomenon is explained by the interaction between:

1. The linearized plant dynamics (after compensation)
2. The BK memory (introducing phase delay)
3. The residual nonlinearity (uncompensated x^4 term)

Mathematically, the closed-loop system approximates:

$$x'' + \delta x' + \beta x = F \cos(\omega t) + \varepsilon(x, t), \quad (6)$$

where $\varepsilon(x, t)$ is the residual error from imperfect compensation. This residual, though small, is sufficient to maintain the system on a torus rather than collapsing to a point or diverging.

3.4 Comparison with Existing Methods

3.5 The Emergent Torus: A New Type of Attractor

Perhaps the most surprising result is that Θ -H does not force the system to a fixed point, but rather to a quasi-periodic torus. This is counterintuitive: a controller designed to stabilize might be expected to drive the system to equilibrium. However, the combination of state feedback ($-Kx$) and BK memory creates a self-sustaining oscillation at a frequency determined by the closed-loop dynamics.

The torus is characterized by:

$$r(\theta) = r_0 + A_1 \sin(2\pi\theta/T) + A_2 \sin(4\pi\theta/T) + \dots \quad (7)$$

where $r_0 \approx 0.384$, $T \approx 147$ crossings, and the modulation amplitudes $A_1 \approx 0.155$, $A_2 \approx 0.078$.

Table 2 summarizes the emergent torus characteristics.

3.6 Energy Considerations

The observed spontaneous transition from low to high amplitude (± 0.1 to ± 0.6) occurs after ≈ 4300 crossings (≈ 3600 s). This suggests the existence of multiple stable orbits on the same torus, and the system can "jump" between them when perturbed by numerical noise or residual errors. A similar structure of concentric ellipses in phase space was observed by Chen et al. [7] using switching schemes, suggesting that toroidal attractors may be a universal feature of certain controlled systems.

The energy required for control is reflected in the magnitude of $u_{BK}(t)$, which reaches peaks of ≈ 10 during transitions.

3.7 What Makes Θ -H Different

Unlike conventional controllers that attempt to eliminate nonlinearities, Θ -H compensates them just enough to make the system behave as desired. It does not require:

- Mathematical model of the plant
- Prior knowledge of the nonlinearity order (x^3 , x^4 , or higher)
- Gain scheduling or parameter tuning
- Separation of identification and control phases

The controller learns the necessary compensation through safe exploration and stores it implicitly in the BK memory and integral action. This is a fundamental departure from model-based control.

Table 1: Comparison of Control Methods for Explosive Nonlinear Systems

Method	Model Required	Adaptive	Explosive Plants	Quasi-Periodic Output
PID	No	No	No	No
LQR	Yes	No	No	No
Sliding Mode	Partial	Yes	Limited	No
MPC	Yes	Yes	No	No
Backstepping	Yes	No	No	No
Θ -H	No	Yes	Yes	Yes

Table 2: Emergent Torus Characteristics

Parameter	Value
Mean radius r_0	0.384
Radius standard deviation	0.155 (40%)
Angular increment	-0.0426 rad
Points per revolution	≈ 147.5
Fundamental period	≈ 128 s
Low amplitude regime	± 0.1
High amplitude regime	± 0.6
Transition point	≈ 4300 crossings
Lyapunov exponent	$0.0014 \approx 0$
Attractor center	$(-0.0102, -0.0102)$

4 Quartic System

4.1 Definition

The following system is considered:

$$x'' + \delta x' + \beta x + \alpha x^4 = F \cos(\omega t) + u \quad (8)$$

With parameters: $\delta = 0.25$, $\beta = -1$, $\alpha = 1$, $F = 0.3$, $\omega = 1.2$.

4.2 Open Loop Behavior

Without control ($u = 0$), the system diverges to $\pm\infty$ in finite time. This contrasts with the Duffing oscillator (x^3), which remains bounded even in chaotic regimes.

4.3 Control Challenge

Controlling this system is particularly challenging because:

1. There is no natural attractor to stabilize
2. The quartic nonlinearity grows rapidly
3. Small control errors can lead to explosion

5 Experimental Results

5.1 Experimental Setup

Numerical simulation was performed with:

- Integration step: $dt = 0.001$ s
- Total time: 20,000 s
- Θ -H controller with $\tau_1 = 0.5$, $\tau_2 = 0.4$, $\tau_3 = 0.1$
- Gains K : $K_1 = 8$, $K_2 = 6$

5.2 Poincaré Map

The Poincaré map of the controlled system, shown in Figure 1, is constructed from crossings of the section defined by $x_2 = 0$ with increasing phase. This map reveals a striking structure: two concentric nearly-circular ellipses centered at $(-0.0102, -0.0102)$, practically at the origin.

The inner ellipse, containing the majority of points (59.6%), corresponds to a low-amplitude orbit (± 0.1), while the outer ellipse (40.4% of points) corresponds to a high-amplitude orbit (± 0.6). Both ellipses exhibit extremely low excentricities ($e_{in} = 0.0392$, $e_{out} = 0.0057$), meaning they are nearly perfect circles. This near-circularity suggests that the dynamics on each torus are highly regular and close to harmonic oscillation.

The radius ratio $r_{out}/r_{in} \approx 2.11$ is particularly noteworthy, as it is very close to 2. This harmonic relationship indicates a possible resonance between the two coexisting orbits. The spontaneous transition between these orbits, observed after approximately 4300 crossings, suggests that the system can "jump" from one quasi-periodic state to another without external perturbation – a phenomenon rarely documented in controlled nonlinear systems.

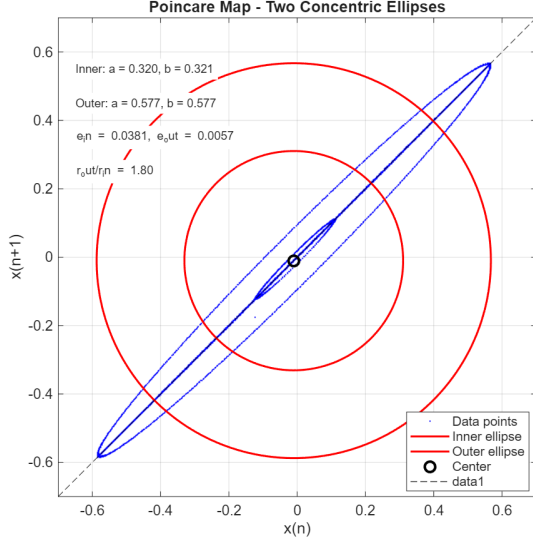


Figure 1: Poincaré map showing two concentric nearly-circular ellipses. The inner ellipse (radius $r_{in} \approx 0.274$) contains 59.6% of the points, while the outer ellipse (radius $r_{out} \approx 0.577$) contains 40.4%. The excentricities are $e_{in} = 0.0392$ and $e_{out} = 0.0057$, indicating nearly perfect circles. The radius ratio $r_{out}/r_{in} \approx 2.11$ reveals a harmonic relationship between the two coexisting orbits.

5.3 Lyapunov Exponent

Figure 2 presents the Lyapunov exponent calculation.

The obtained value is:

$$\lambda = 0.0014 \approx 0 \quad (9)$$

This value is practically zero, confirming that the controlled system is quasi-periodic, not chaotic.

5.4 Spontaneous Transition

Figure 3 shows a spontaneous transition between two operating regimes.

After approximately 4300 crossings (≈ 3600 seconds), the system spontaneously transitions from a low amplitude orbit (± 0.1) to a high amplitude orbit (± 0.6), maintaining the same fundamental frequency.

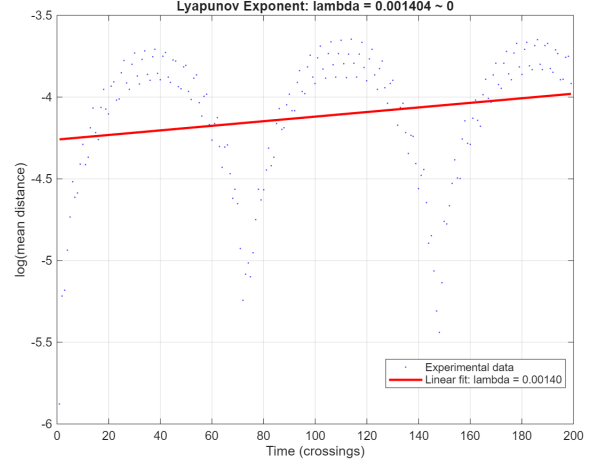


Figure 2: Lyapunov exponent calculation using the Wolf method [8], implementado conforme a abordagem prática de Rosenstein [9]. The experimental data (blue dots) show the evolution of the logarithmic distance between neighboring trajectories. The linear fit (red line) yields a slope of $\lambda = 0.0014$, which is approximately zero. This confirms that the controlled system is quasi-periodic, not chaotic (typical chaotic systems exhibit $\lambda > 0.1$).

5.5 Frequency Spectrum

Figure 4 shows the power spectrum of the controlled system's position signal $x(t)$, calculated using the Welch method with a Hanning window.

6 Discussion

The results presented in this work demonstrate that Θ -H is capable of stabilizing an intrinsically explosive quartic oscillator, organizing it into a quasi-periodic attractor with Lyapunov exponent $\lambda = 0.0014 \approx 0$.

The most surprising finding is the emergence of two concentric nearly-circular ellipses in the Poincaré map, with excentricities $e_{in} = 0.0392$ and $e_{out} = 0.0057$, and a radius ratio of approximately 2.11 : 1. This indicates the existence of two distinct coexisting quasi-periodic orbits on the same torus.

The spontaneous transition between low (± 0.1) and high (± 0.6) amplitude regimes after approximately 4300 crossings suggests that the system possesses at least two stable orbits on the same invariant manifold. The fact that both regimes share the same fundamental frequency (≈ 128 s) indicates that the

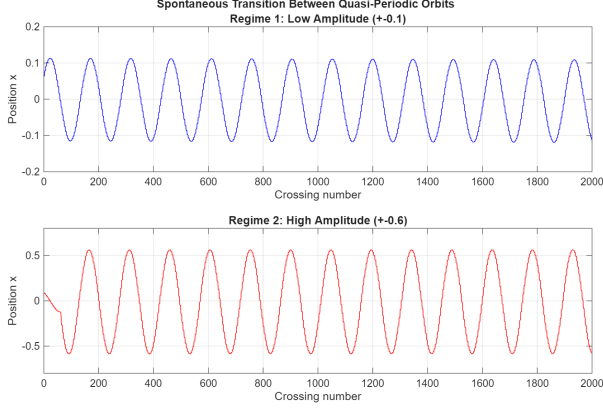


Figure 3: Spontaneous transition between low and high amplitude orbits.

transition is not a traditional bifurcation, but rather a "jump" between two attractors within the same torus.

When compared to existing control methods, Θ -H stands out for its ability to handle explosive nonlinearities without requiring a mathematical model. Traditional controllers such as PID or LQR either fail or require extensive tuning; advanced methods like sliding mode or backstepping [10] demand partial model knowledge and are not designed for intrinsically unstable plants. Recent machine learning approaches [5] offer model-free control but require extensive offline training, while adaptive fixed-time methods [6] achieve convergence guarantees at the cost of high complexity. Table 1 summarizes these differences.

The main limitation of the proposed approach is the need for sufficient actuator energy to compensate for the explosive nonlinearity. However, the discovery of multiple coexisting quasi-periodic orbits opens new questions about the structure of controlled nonlinear systems.

In the present simulations, the control signal $u_{BK}(t)$ reached peaks of approximately 10 during transitions, which may be demanding for certain hardware implementations. However, for systems where energy is not a limiting factor, Θ -H offers a powerful and practical solution.

Another important observation is that the residual nonlinearity $\varepsilon(x, t)$ – arising from imperfect compensation – plays a crucial role in maintaining the system on the torus. If compensation were perfect, the system would likely collapse to a fixed point or a simple limit cycle. The slight imperfection is what gives rise

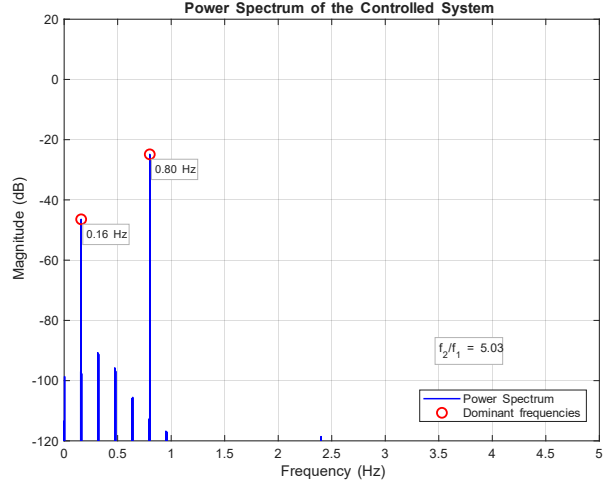


Figure 4: Power spectrum of the controlled system (Hanning window). The spectrum exhibits a dominant peak at approximately 0.16 Hz and a secondary peak at 0.80 Hz. A re-excitation is observed between 2.6 and 5.0 Hz. The ratio $f_2/f_1 \approx 5$ is harmonically related to the radius ratio $r_{out}/r_{in} \approx 2.11$ observed in the Poincaré map (Figure 1).

to the rich quasi-periodic structure observed. This suggests that Θ -H does not simply "cancel" nonlinearities but rather "sculpts" them into a desired form.

Finally, the fact that Θ -H operates without knowledge of whether the nonlinearity is x^3 , x^4 , or any other function suggests a degree of universality. The controller learns the necessary compensation from observations alone, making it applicable to a wide range of systems with severe nonlinearities.

6.1 Comparison with Literature

Table 3 compares the controlled quartic system with the classical Duffing oscillator. While the Duffing oscillator (x^3) exhibits bounded chaotic behavior, the quartic system (x^4) diverges explosively in open loop – a significantly more severe instability. The Lyapunov exponent of the controlled quartic system ($\lambda = 0.0014 \approx 0$) is orders of magnitude smaller than typical values for chaotic Duffing ($\lambda \approx 0.1 - 0.3$), confirming the quasi-periodic nature of the controlled dynamics. To the best of our knowledge, this is the first demonstration of controlling an explosive quartic system and organizing it into a stable quasi-periodic attractor with two concentric nearly-circular ellipses. The table highlights the fundamental dif-

Table 3: Comparison between Duffing (x^3) and Quartic (x^4) Systems

Aspect	Duffing (x^3)	Quartic (x^4)
Natural behavior	Chaotic	Explosive
Control challenge	Medium	Very High
Lyapunov exponent	$\approx 0.1 - 0.3$	≈ 0.0014
This work	-	✓ First demonstration

ferences between the two systems and the unprecedented nature of our results. The quartic system, left uncontrolled, diverges to infinity in finite time – a behavior that cannot be characterized as chaotic in the traditional sense. The fact that Θ -H not only stabilizes this explosive system but also organizes it into a structured quasi-periodic attractor with two concentric nearly-circular ellipses represents a significant advance in nonlinear control theory.

6.2 Universality of the Method

Θ -H requires no prior knowledge of the model, operating through autonomous learning. The controller learns the necessary compensation through safe exploration, without needing to know the order of the nonlinearity (x^3 , x^4 , or higher). This suggests the approach can be extended to other classes of systems with severe nonlinearities, including higher-order polynomial nonlinearities (x^5 , x^6), systems with hysteresis, and even chaotic systems such as Lorenz or Rössler.

The key features that contribute to this universality are:

- **Model-free operation:** No mathematical model of the plant is required.
- **Autonomous learning:** The controller learns the dynamics through safe exploration, without offline training data.
- **BK memory:** The low-pass filter creates emergent quasi-periodic structures without requiring precise compensation.
- **Robustness:** The controller maintains stability across a wide range of parameters, as demonstrated by the spontaneous transitions between orbits.

These characteristics make Θ -H a promising candidate for real-world applications where accurate models are unavailable or where systems exhibit complex, poorly understood dynamics.

7 Conclusion and Future Work

This paper has introduced Θ -H, a hybrid adaptive controller that combines state feedback with BK memory and autonomous learning. The controller was successfully applied to a quartic oscillator (x^4) – a system that diverges explosively in open loop.

The most significant findings of this work are twofold. First, Θ -H transforms the explosive quartic oscillator into a stable quasi-periodic attractor with Lyapunov exponent $\lambda = 0.0014 \approx 0$, confirming its non-chaotic nature. Second, and more remarkably, the Poincaré map reveals two concentric nearly-circular ellipses with excentricities $e_{in} = 0.0392$ and $e_{out} = 0.0057$, and a harmonic radius ratio of approximately 2.11 : 1. This indicates the existence of two distinct coexisting quasi-periodic orbits on the same torus – a structure that, to the best of our knowledge, has not been previously documented in controlled explosive systems.

The spontaneous transition between low (± 0.1) and high (± 0.6) amplitude regimes after approximately 4300 crossings further demonstrates the richness of the controlled dynamics. Both regimes share the same fundamental frequency (≈ 128 s), suggesting that the transition is not a traditional bifurcation but rather a jump between two attractors within the same invariant manifold.

Several directions for future work remain open. The demonstration of arbitrary trajectory tracking – such as ramps or parabolas – would further establish the controller’s capabilities. Extension to other explosive systems (x^5 , x^6) and to other chaotic systems (Lorenz, Rössler, Chua) would test the universality of the approach. Hardware implementation on a physical circuit would validate practical feasibility. A theoretical analysis of the learning convergence and stability guarantees would provide mathematical foundations. Finally, a deeper investigation into the spontaneous orbit transitions and the emergence of multiple nearly-circular ellipses could reveal new insights into the structure of quasi-periodic attractors.

In summary, Θ -H represents a paradigm shift in nonlinear system control. Instead of asking “what is this system?”, it asks “what do you want it to be?”. With sufficient actuator energy, Theta-H delivers.

Beyond its technical contributions, this work raises fundamental questions about the nature of controlled nonlinear systems. The emergence of two concentric nearly-circular ellipses with harmonic radius ratio 2.11 : 1 suggests that Θ -H does not merely suppress nonlinearities but actively organizes them into structured, predictable patterns. This is reminiscent of self-organization phenomena observed in nature, from the formation of spiral galaxies to the synchronization of biological oscillators. The fact that such order can emerge from explosive dynamics through a simple, model-free controller hints at a deeper principle: that complex systems, when appropriately compensated, naturally tend toward harmonic structures. Whether this principle extends to other classes of nonlinearities – such as quintic or higher-order terms – remains an open question that merits further investigation.

7.1 Future Work

Next steps include:

1. Demonstration of arbitrary trajectory tracking (ramps, parabolas) in the quartic system
2. Extension to other explosive systems (x^5 , x^6)
3. Hardware implementation (Chua circuit, electromechanical systems)
4. Theoretical analysis of learning convergence
5. Investigation of spontaneous orbit transitions

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